Low-Dropout (LDO) Voltage Regulator in SKY130A Simulation, Physical Verification & Post-Layout Validation

Silicluster v3 Analog Project Submission | VDD = 1.8 V | VOUT = 1.2 V | Iload = 0 – 5 mA

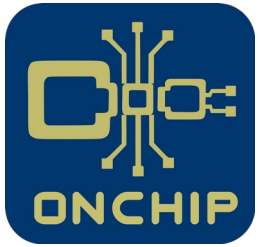
Luigy Amado Jeison Acevedo Daniel Castellanos

Integrated Circuit and Systems Research Group – OnChip

Universidad Industrial de Santander, Bucaramanga - Colombia

lduran2805@gmail.com | luigy2230397@correo.uis.edu.co





Outline

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- 2. Top-Level LDO Regulator Schematic Architecture (LDO.sch)
- 3. VDD-Independent Self-Biased Current Reference Circuit (self_bias.sch)
- 4. Two-Stage Miller-Compensated Error Amplifier (OTA.sch)
- 5. Pre-Layout Simulation Results: AC Stability & PSRR (225 PVT Corners)
- 6. Pre-Layout Simulation Results: Transient Response & DC Regulations
- 7. Monte Carlo Statistical Analysis (Process & Mismatch Variability)
- 8. Physical Layout Design, Area (152.4 µm × 82.9 µm) & M4/M5 Power Grid (LDO.gds)
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- 10. Post-Layout (PEX) Validation: Response Curves Across PVT Corners
- 11. Final Comparison: Pre-Layout vs. Post-Layout (PEX)

1. Specifications & External Interface (Silicluster v3)

- PDK Technology: SkyWater 130 nm Open Source PDK (sky130A).
- Nominal Analog Supply: VDD = 1.8 V (range: 1.62 V – 1.98 V, VSS = 0 V).
- Input Voltage Range: VIN = 1.3-1.8 V nominal (mean Dropout ≈ 60.7 mV).
- Reference & Output: External Vref = 0.6 V → VOUT = 1.2 V (Iload = 0 to 5 mA).
- Total Physical Area: 152.43 µm × 82.91 µm (standard limit: 180 µm × 115 µm).
- Power Grid: Vertical tracks on Metal 4 (M4) and horizontal rails on Metal 5 (M5).
- Power Grid: Vertical tracks on Metal 4 (M4) and horizontal rails on Metal 5 (M5).
- Physical Verification: 0 DRC errors (sky130_drc.txt) and 100% clean LVS (LDO.lvsdb).

Pin Name	Direction	Layout Layer	Track Width	Requested Chip Assignment
VDD	inout	met4 (V) / met5 (H)	1.0 µm / 2.0 µm	Power Grid VDD (1.8 V)
VSS	inout	met4 (V) / met5 (H)	1.0 µm / 2.0 µm	Power Grid VSS (0 V)
VIN	input	met3 (70/20)	2.0 µm	1 Analog Input Pad (1.3–1.8 V)
Vref	input	met3 (70/20)	0.30 µm	1 Analog Reference Pad (0.6 V)
VOUT	output	met3 (70/20)	2.0 µm (x2 branches)	2 Analog Output Pins / Pads (connected to the two met3 branches)

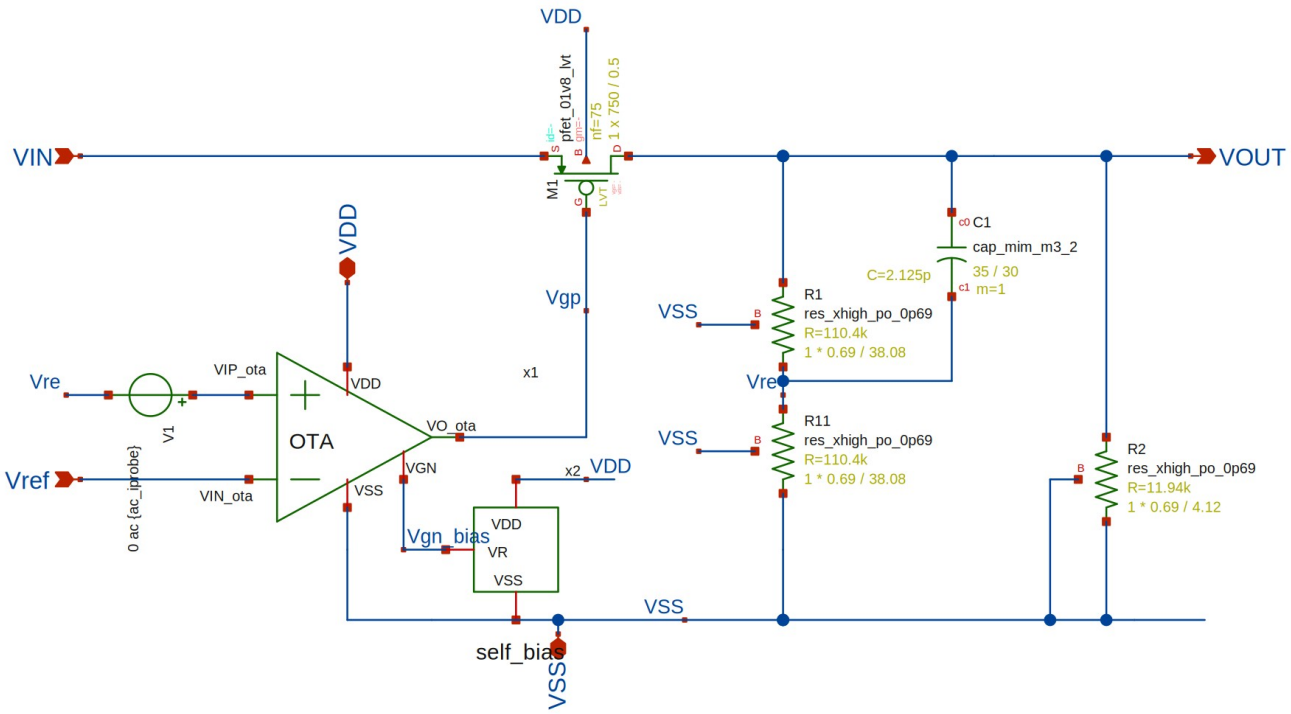
External Verilog Interface (LDO.v):

```
module LDO (
    inout  VDD,   inout  VSS,
    input  VIN,   input  Vref,
    output VOUT
);
endmodule
```

2. Top-Level LDO Schematic Architecture

- The LDO regulator integrates 3 core building blocks in closed loop:
- 1) Self-Bias Generator (x2): Provides bias voltage Vgn_bias (~0.71 V) to the OTA without external current sources.
- 2) OTA Error Amplifier (x1): Compares Vref (0.6 V) against feedback node Vre and drives pass transistor gate Vgp.
- 3) Power & Feedback Stage: PMOS LVT M1 (W=750 µm, L=0.5 µm), divider R1=R11=110.4 kΩ (VOUT=1.2 V), R2=11.94 kΩ, and MIM cap C1=2.1 pF.

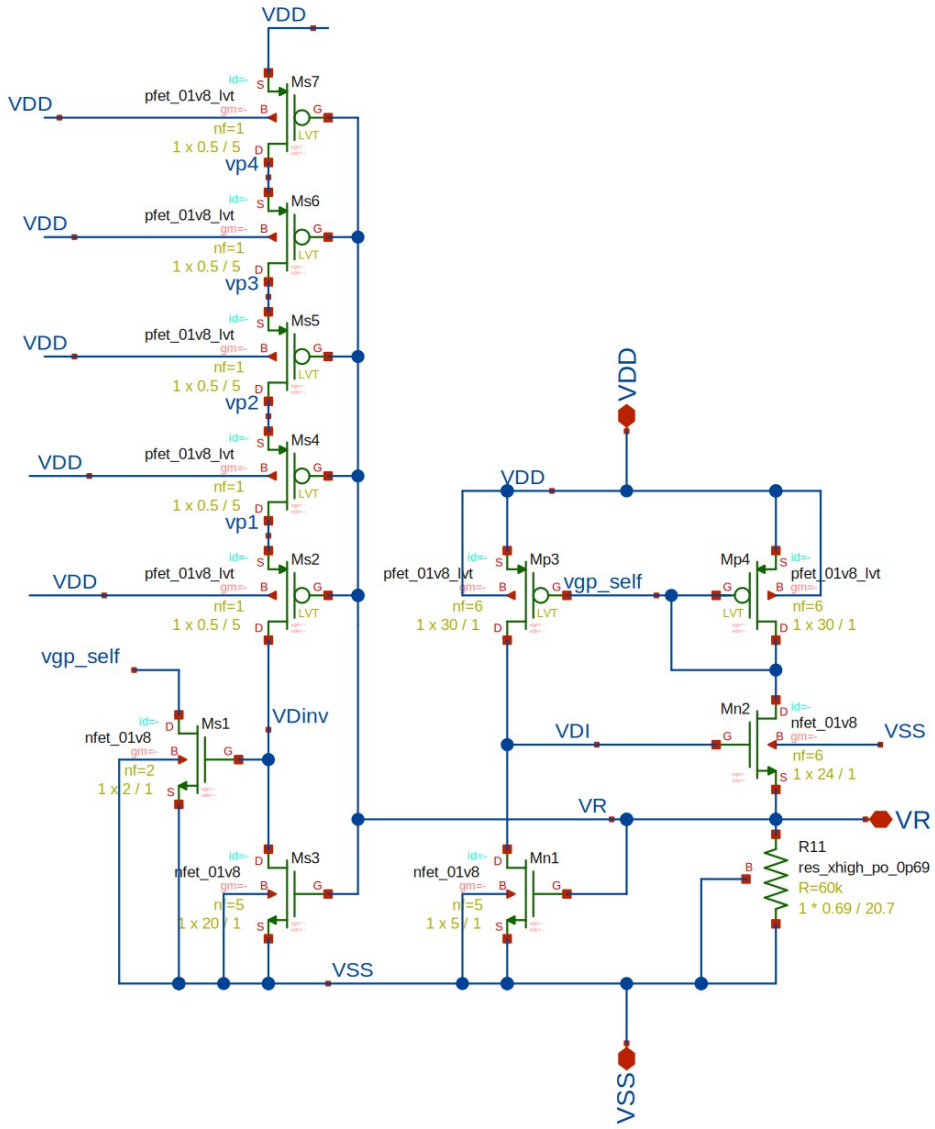
Elem.	Sky130A Model	Dimensions	Function
M1	pfet_01v8_lvt	W=750µm, L=0.5µm, nf=75	PMOS pass transistor
R1, R11	res_xhigh_po_0p69	0.69×38.08µm (110.4 kΩ)	Feedback divider (Vre = 0.5·VOUT)
R2	res_xhigh_po_0p69	0.69×4.12µm (11.94 kΩ)	Minimum bleeder load (~100 µA)
C1	cap_mim_m3_2 (M4)	35µm × 30µm (2.10 pF)	Feedforward capacitor



3. Self-Biased Current Reference Generator (Self-Bias)

- Beta-Multiplier topology with source resistive degeneration (R11 = 60 kΩ).
- Bias Core: 1:1 PMOS LVT current mirror (Mp3, Mp4 with W=30 µm, L=1 µm, nf=6) and NMOS pair (Mn1 with W=5 µm, Mn2 with W=24 µm).
- Nominal Operating Point (TT, 27 °C, 1.8 V): VR = 0.7138 V and Ibias = 11.82 µA.
- Start-Up Circuit: 5 series PMOS LVT devices (Ms2, Ms4–Ms7 with W=0.5 µm, L=5 µm) with Ms3 and Ms1 preventing the zero-current state at power-up.

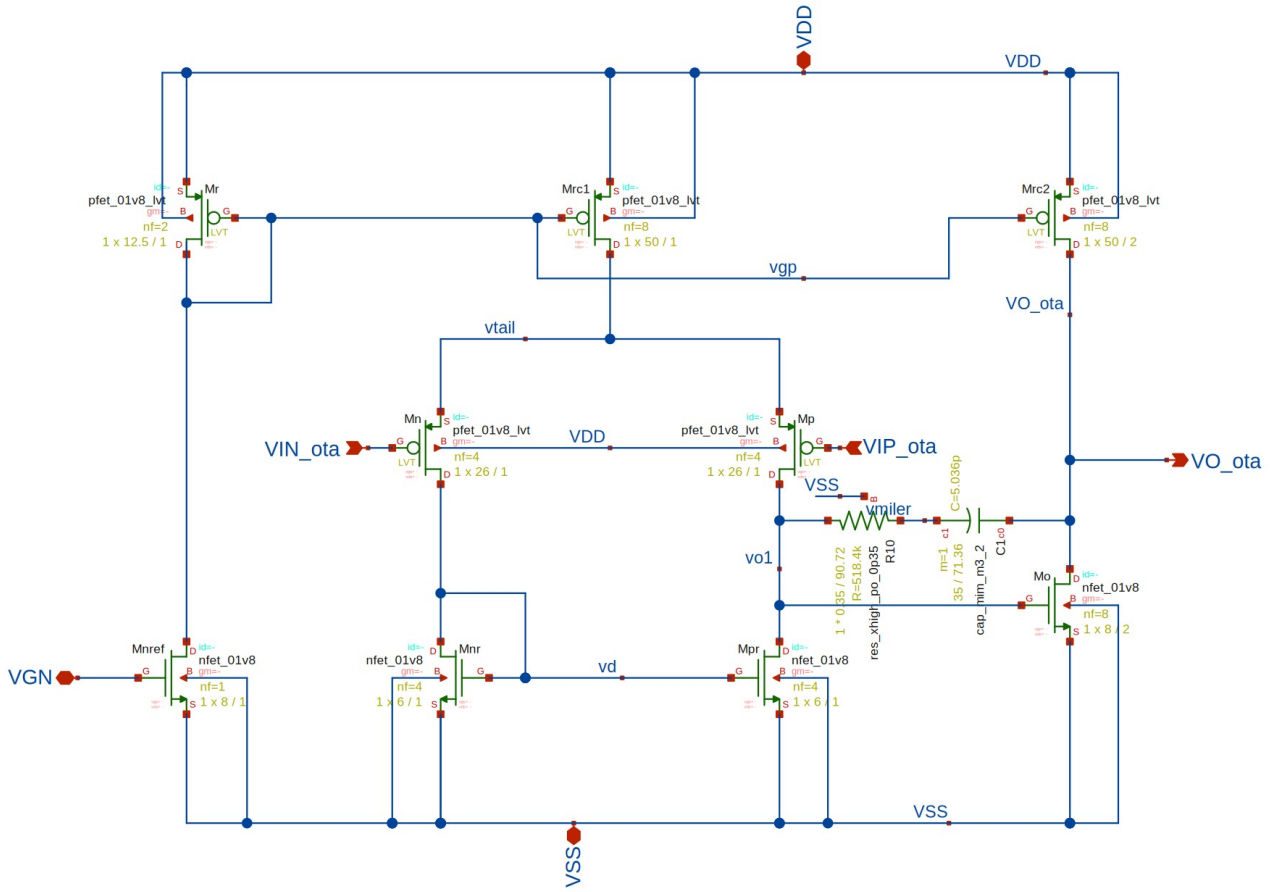
Device	Sky130A Model	W [µm]	L [µm]	nf	Role in Circuit
Mp3, Mp4	pfet_01v8_lvt	30.0	1.0	6	Upper PMOS mirror
Mn1	nfet_01v8	5.0	1.0	5	Left NMOS reference
Mn2	nfet_01v8	24.0	1.0	6	R11-degenerated NMOS
R11	res_xhigh_po_0p69	0.69	20.7	1	Ibias resistor (60 kΩ)
Ms3, Ms2 ,Ms4–7	pfet_01v8_lvt	0.5	5.0	1	Start-up branch
Ms1	nfet_01v8	2 / 20	1.0	2 / 5	Start-up injection & cutoff



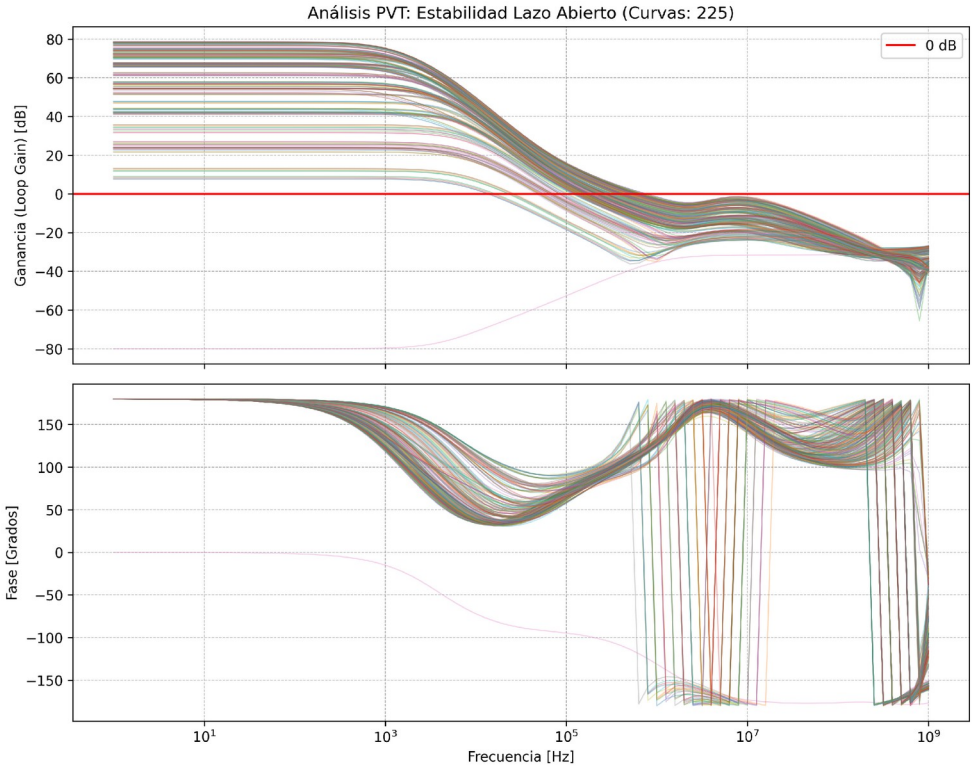
4. Two-Stage OTA Error Amplifier (OTA.sch)

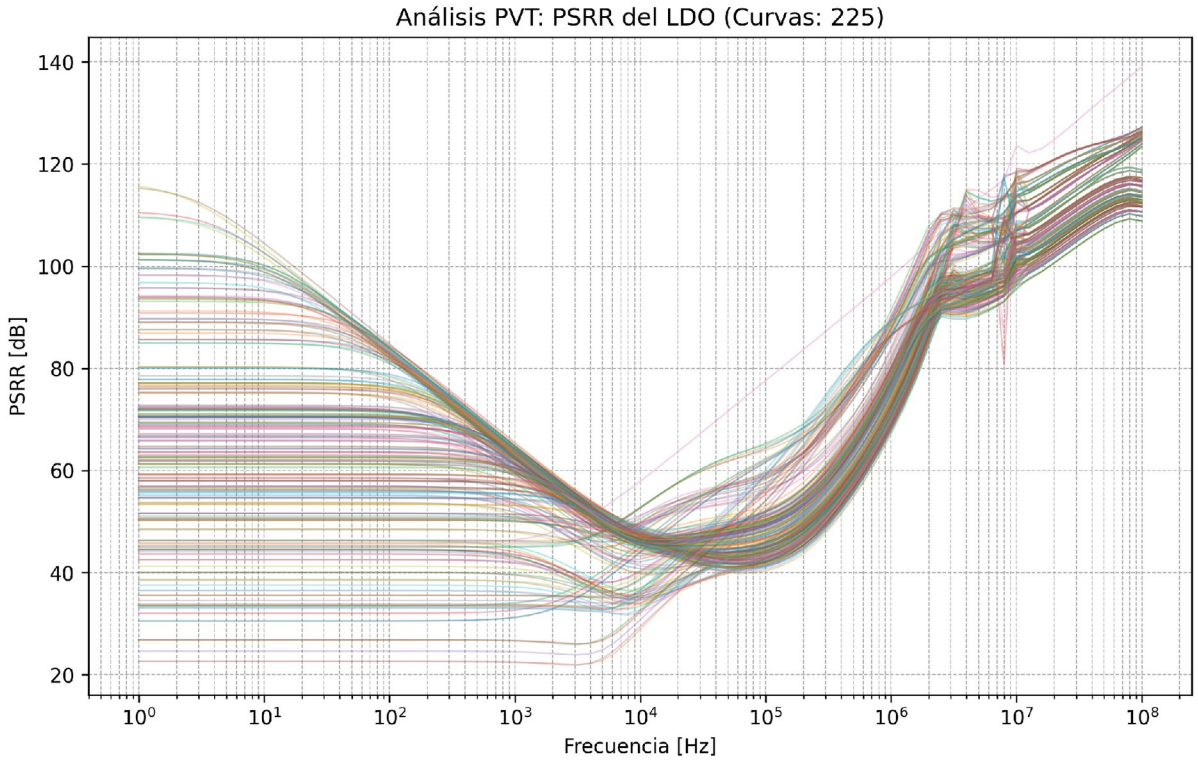
- Two-stage operational transconductance amplifier driving pass transistor M1 gate.
- First Stage: PMOS LVT differential pair (Mn, Mp with W=26 µm, L=1 µm, nf=4) with NMOS active load mirror (Mnr, Mpr with W=6 µm, L=1 µm, nf=4).
- Second Stage: Common-source NMOS stage (Mo with W=8 µm, L=2 µm, nf=8) with PMOS active load (Mrc2 with W=50 µm, L=2 µm, nf=8).
- Miller Compensation: MIM capacitor C1 = 5.0 pF (44.6 µm × 56 µm) and nulling resistor R10 = 518.4 kΩ.

Device	Sky130A Model	W [µm]	L [µm]	nf	Function in OTA
Mn, Mp	pfet_01v8_lvt	26.0	1.0	4	Input differential pair
Mnr, Mpr	nfet_01v8	6.0	1.0	4	Active load mirror
Mrc1 / Mr	pfet_01v8_lvt	50 / 12.5	1.0	8 / 2	Tail current source & mirror
Mrc2 / Mo	pfet_lvt / nfet	50 / 8.0	2.0	8 / 8	Second-stage output
Mnref	nfet_01v8	8.0	1.0	1	VGN bias sink
C1 / R10	cap_mim / res_po	44.6×56	-	1	Miller network (5pF + 518kΩ)



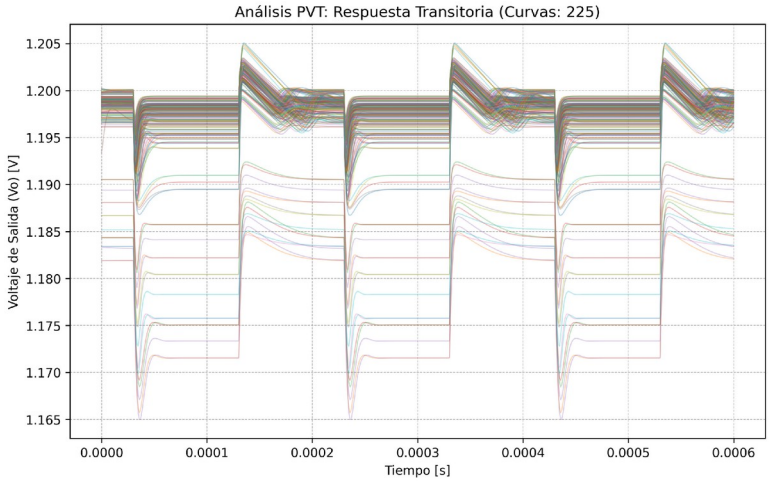
5. Pre-Layout Simulation: AC Stability & PSRR (PVT)

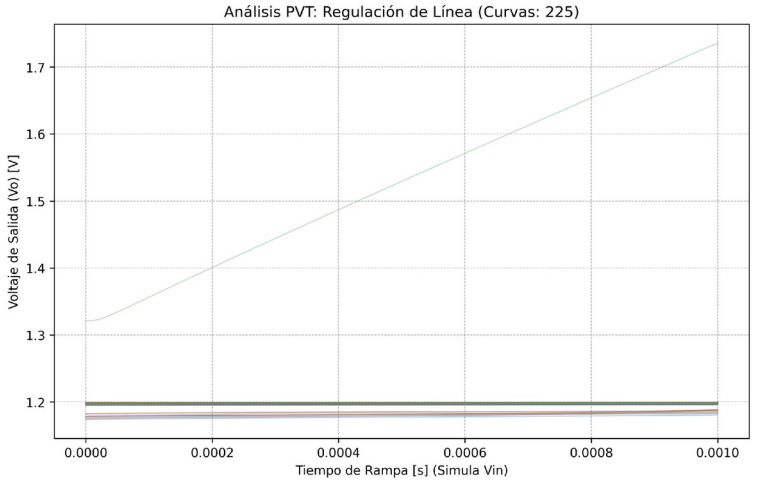


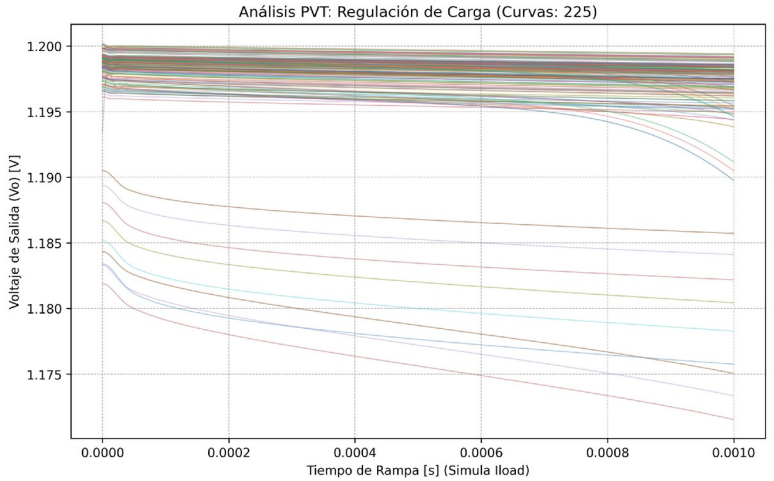


- Sweep across 225 PVT corners: T ∈ {-40 °C, 27 °C, 125 °C}, VDD ∈ {1.62 V, 1.80 V, 1.98 V}, and 25 Sky130 process corners.
- Open-Loop Stability (Iload = 5 mA): Mean Phase Margin = 92.69° (min = 64.18°, max = 123.50°); Mean UGB = 275.3 kHz.
- Power Supply Rejection Ratio (PSRR): Mean of 64.90 dB @ 10 Hz, 62.36 dB @ 1 kHz, 55.49 dB @ 10 kHz, and 48.46 dB @ 1 MHz.

6. Pre-Layout Simulation: Transient & Regulations

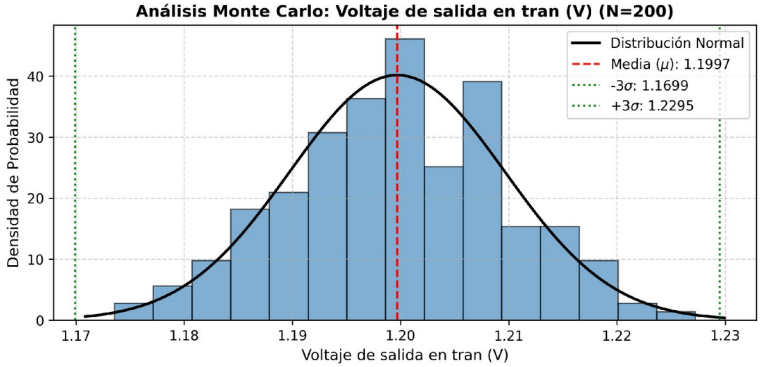


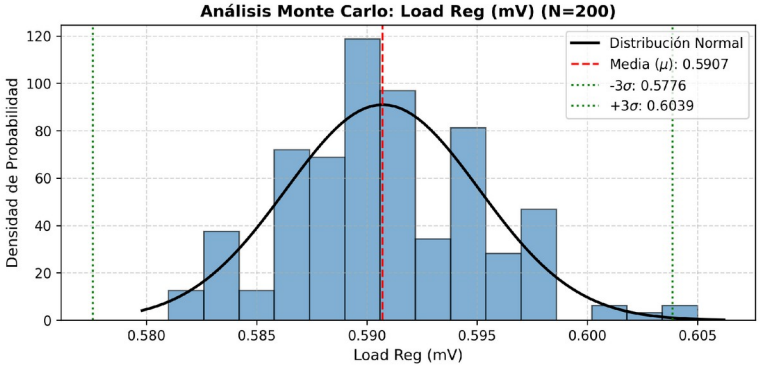


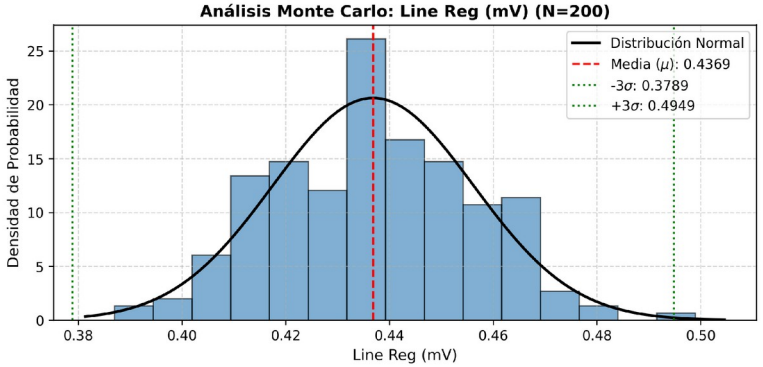


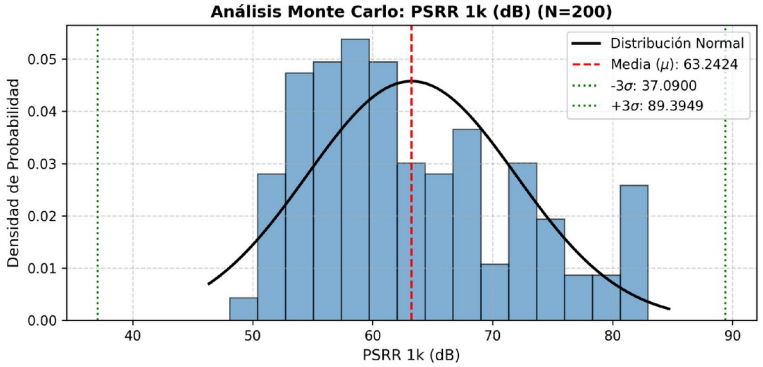
Pre-Layout Metric (225 PVT Corners, Iload = 5 mA)	Minimum	Mean	Maximum	Unit
Transient Output Voltage (Vo_tran)	1.1776	1.1972	1.1998	V
Undershoot on 1 µA → 5 mA step (tr = 100 ns)	2.072	3.883	6.669	mV
Overshoot on 5 mA → 1 µA step (tf = 100 ns)	1.000	3.132	5.616	mV
Load Regulation (Iload from 1 µA to 5 mA)	0.602	1.436	8.739	mV
Line Regulation (VIN from 1.4 V to 1.8 V)	0.286	2.631	11.50	mV
Dropout Voltage (at 90% of Vo_nom)	34.33	58.15	84.79	mV

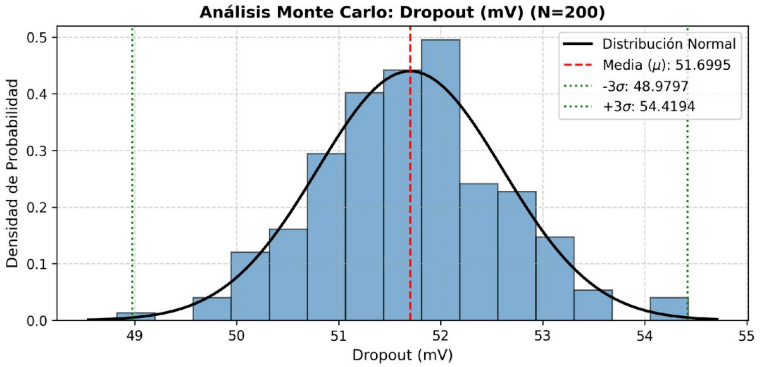
7. Monte Carlo Statistical Analysis (tt_mm, Iload = 5 mA)

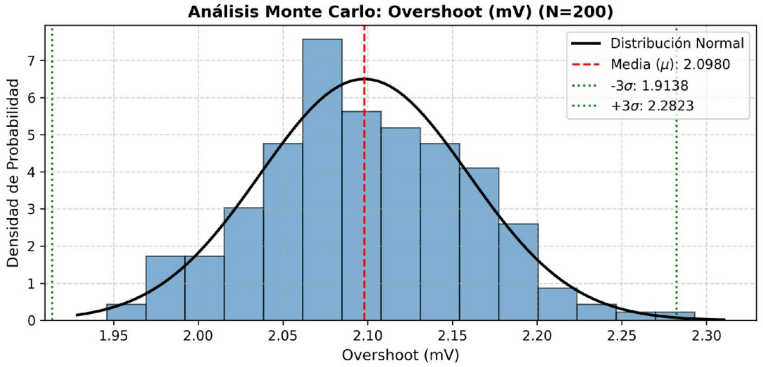




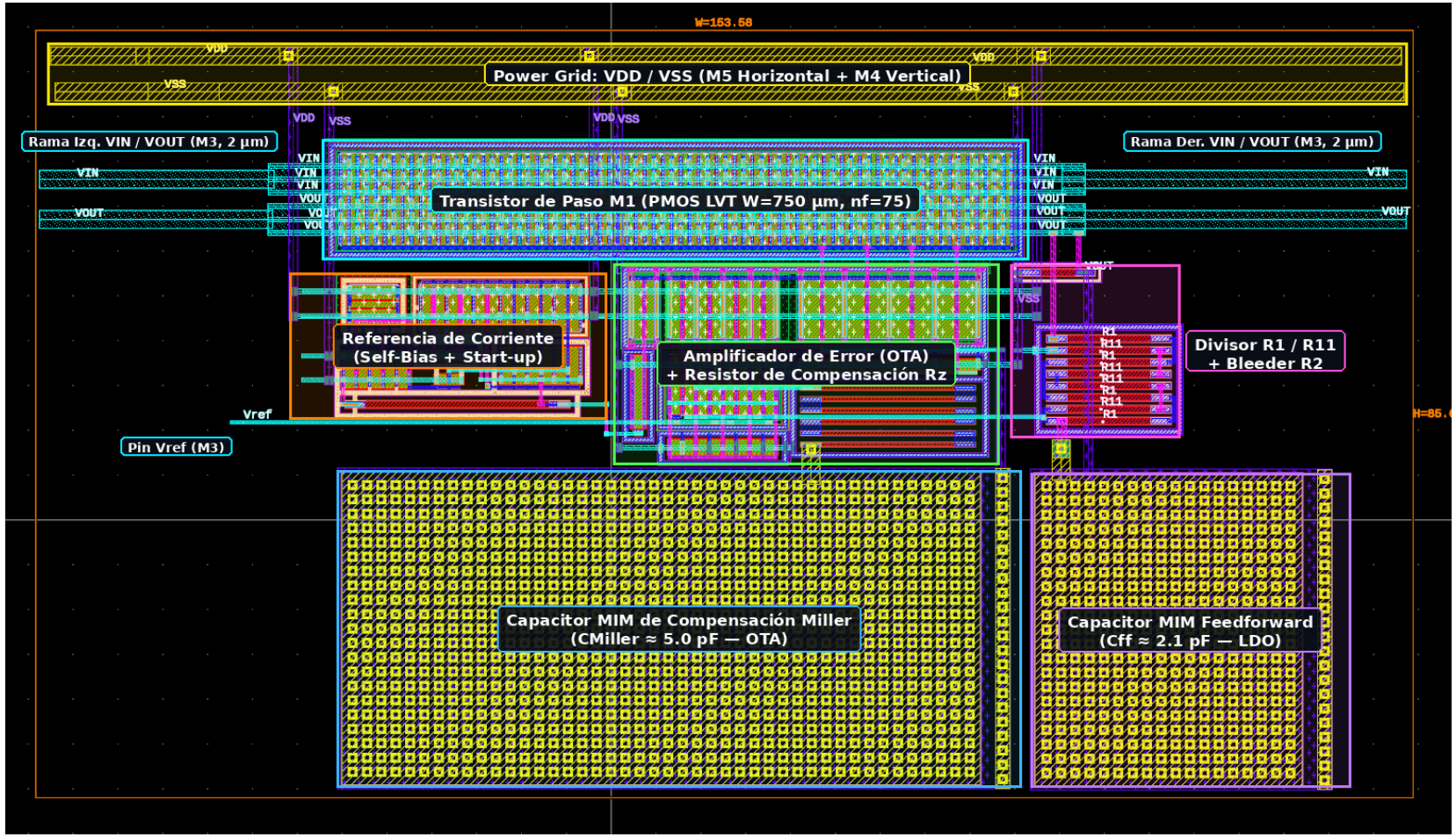






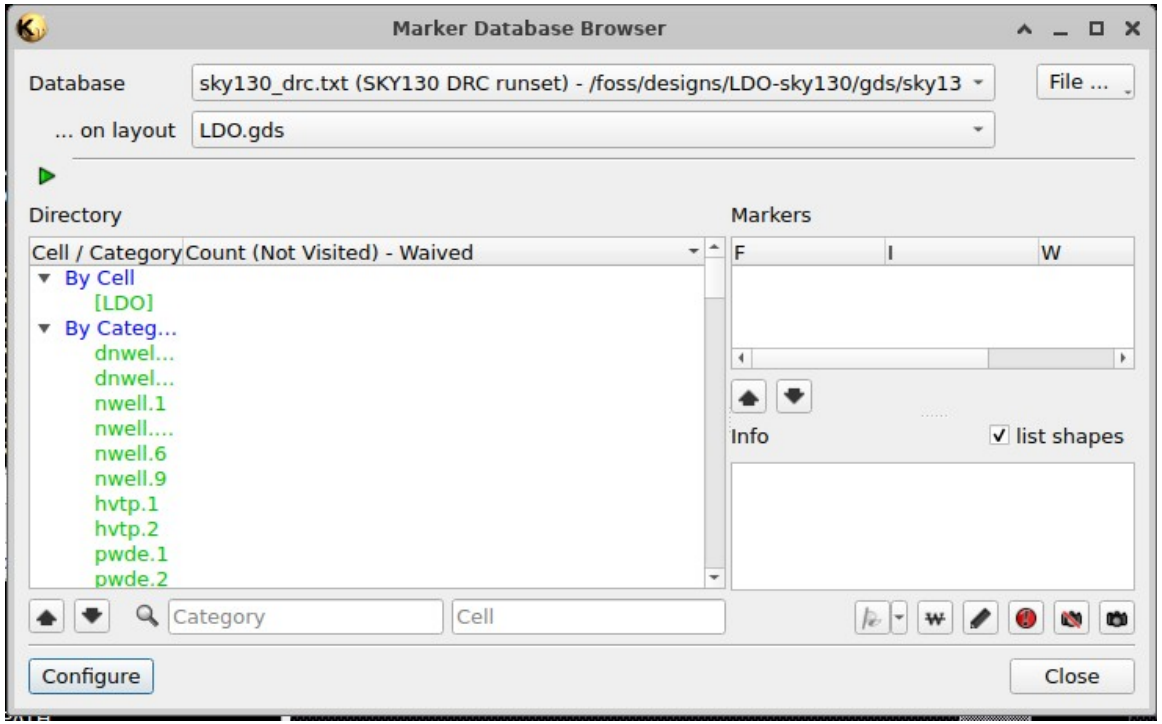


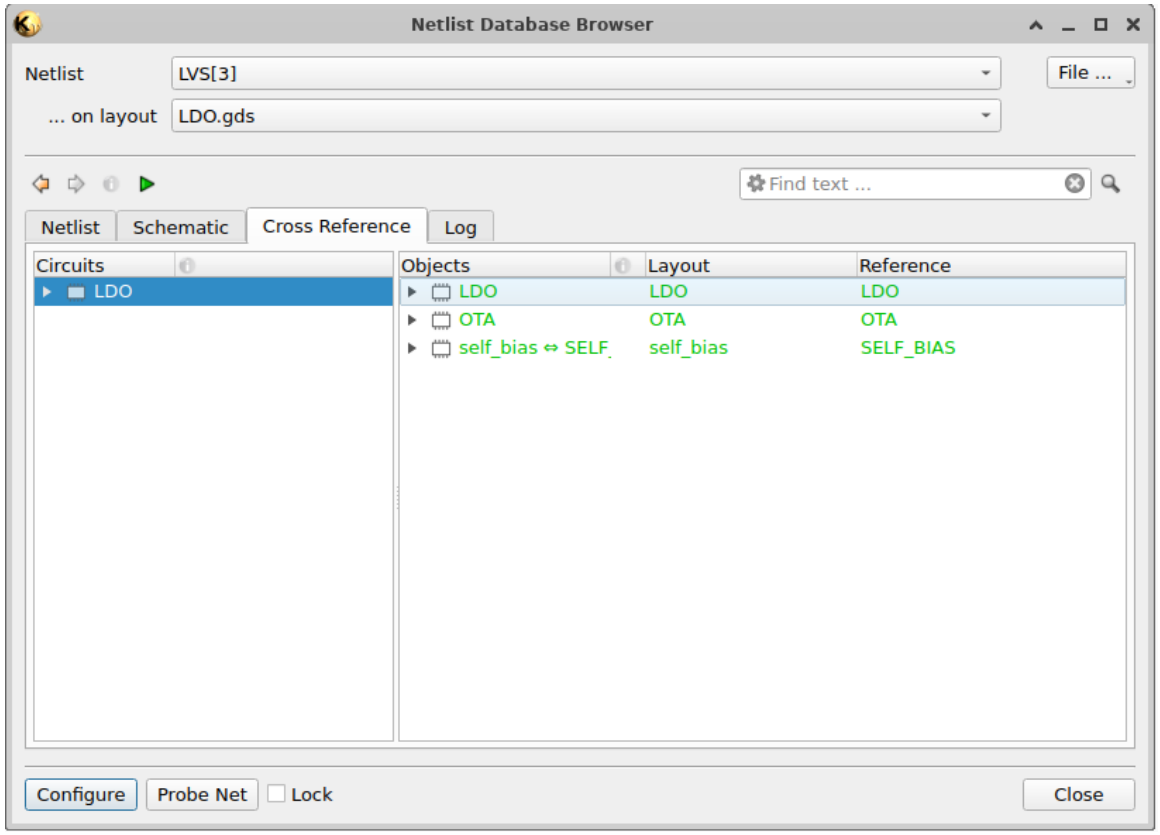
8. Physical Layout Design & M4/M5 Power Grid (LDO.gds)



- Total Dimensions: Width = 152.43 µm (≤ 180 µm), Height = 82.91 µm (≤ 115 µm), Area = 0.0126 mm².
- Power Grid Compliance (Section 11): Vertical distribution tracks on Metal 4 (M4) connected to horizontal main VDD and VSS rails on Metal 5 (M5).
- Matching & Symmetry: OTA differential pair and current mirrors surrounded by dedicated guard rings for latch-up and substrate noise immunity.
- Pass Transistor M1: 75-finger interdigitated array (Wtotal = 750 µm) in the upper region with dense M1–M3 vias to minimize IR drop and series resistance.
- MIM Capacitors (M3/M4): Placed in the lower region efficiently utilizing the area beneath top-level routing.

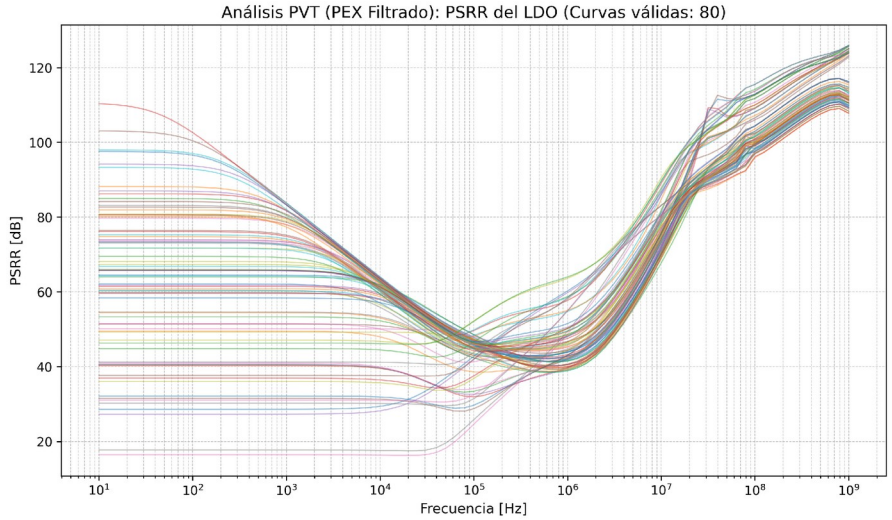
9. Physical Verification Evidence: Clean DRC & LVS

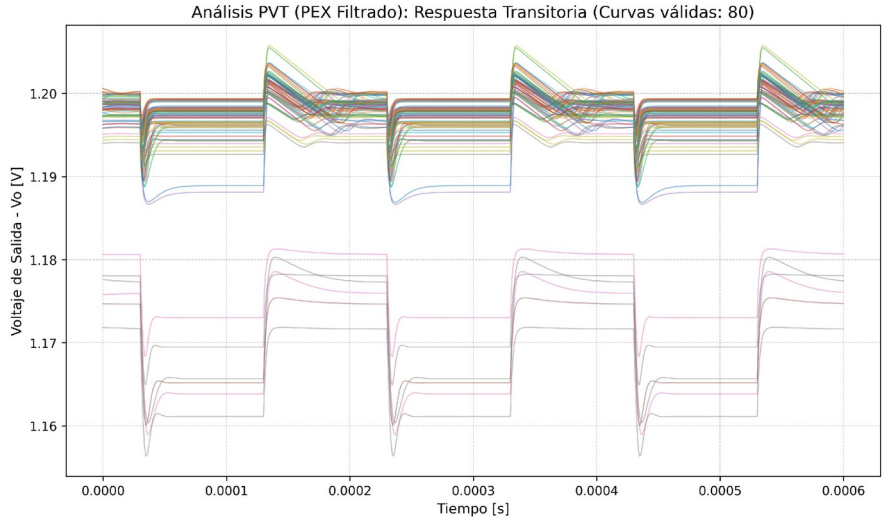


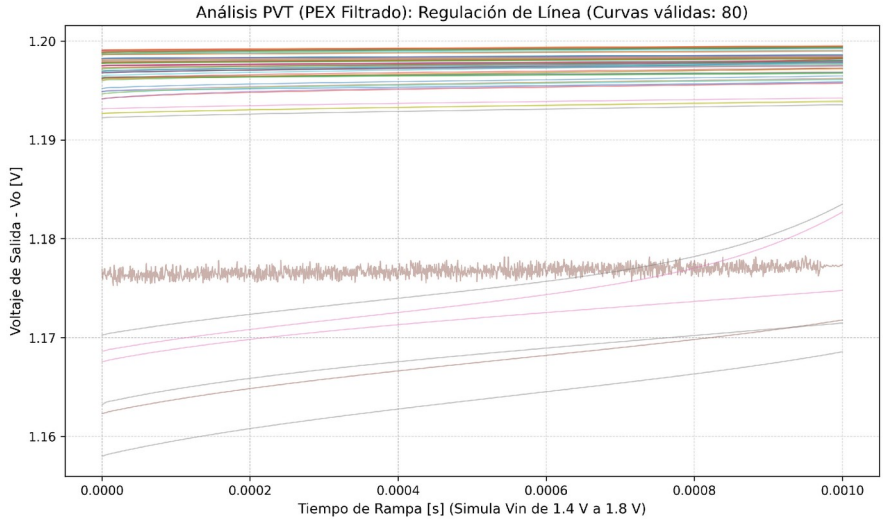


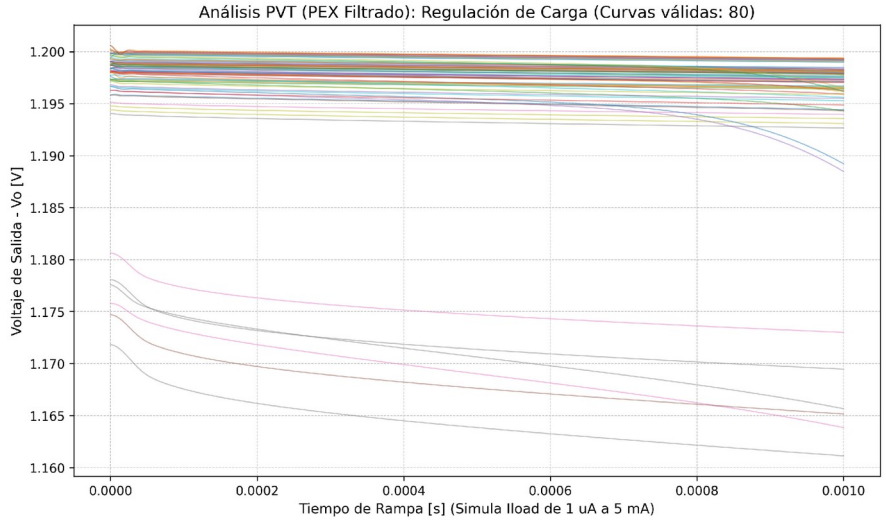
- DRC Verification: Executed in KLayout using the official SKY130 DRC runset on top cell 'LDO'. Result: 0 violations (100% clean).
- LVS Verification (Right - LDO.lvsdb): Executed in KLayout comparing hierarchical layout 'LDO.gds' against Xschem schematic. Result: Exact match (green) across all 3 hierarchical levels (LDO, OTA, and self_bias).

10. Post-Layout (PEX) Validation: PVT Corner Curves









11. Final Comparison: Pre-Layout vs. Post-Layout (PEX)

Parameter / Specification	Pre Min	Pre Mean	Pre Max	PEX Min	PEX Mean	PEX Max	Unit
Settled Output Voltage (Vo_tran)	1.1776	1.1972	1.1998	1.1669	1.1959	1.1999	V
Line Output Voltage (Vo_line)	1.1773	1.1982	1.2000	1.1635	1.1953	1.1993	V
PSRR @ 10 Hz	22.61	64.90	115.65	16.44	63.05	110.32	dB
PSRR @ 1 kHz	22.61	62.36	84.81	16.43	61.15	83.80	dB
PSRR @ 10 kHz	22.45	55.49	65.44	16.37	53.90	64.04	dB
PSRR @ 1 MHz	40.34	48.46	77.63	38.89	46.64	64.00	dB
Undershoot (1 µA → 5 mA)	2.072	3.883	6.669	1.426	3.570	6.938	mV
Overshoot (5 mA → 1 µA)	1.000	3.132	5.616	0.128	2.936	5.451	mV
Line Regulation (1.4 V – 1.8 V)	0.286	2.631	11.50	0.295	1.226	11.60	mV
Load Regulation (1 µA – 5 mA)	0.602	1.436	8.739	0.602	1.673	10.48	mV
Dropout Voltage (Iload = 5 mA)	34.33	58.15	84.79	37.59	60.74	87.65	mV

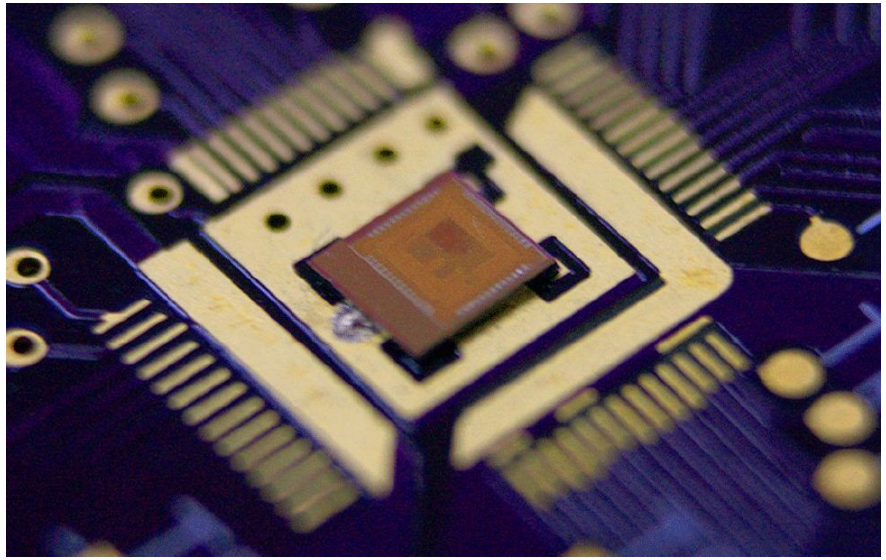
- Validation Conclusion: The full RC-extracted post-layout model (LDO_pex.spice, 35,359 lines) maintains accurate regulation at VOUT = 1.1959 V (< 1.3 mV deviation vs. schematic), mean PSRR > 61 dB up to 1 kHz, and mean Dropout of 60.74 mV.

Thanks

Universidad Industrial de Santander



ONCHIP



lduran2805@gmail.com | luigy2230397@correo.uis.edu.co